

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculator is **allowed**.

1. **Attempt any two of the following:** 12
 - a. Explain the publisher-subscriber communication model.
 - b. How does the web sockets based communication APIs work? Explain.
 - c. How is IoT applied in healthcare? Explain.
 - d. What are the characteristics of IoT? Explain.

 2. **Attempt any two of the following:** 12
 - a. Explain the taxonomy of IoT.
 - b. Compare and contrast IoT Stack and Web Stack.
 - c. Explain the interoperability between IoT devices.
 - d. Explain the assembly, procedural and functional programming paradigms.

 3. **Attempt any two of the following:** 12
 - a. What are different types of virtualization? Explain each.
 - b. What is storage virtualization? Explain its types.
 - c. What are the design challenges in IoT design?
 - d. Explain the data collection and processing cycle.

 4. **Attempt any two of the following:** 12
 - a. Describe how IoT in agriculture and IoT in homes can be adopted to improve the life of citizens in developing countries.
 - b. How is IoT used in politics and military? Explain.
 - c. What are the advantages and disadvantages of the integrated CloudIoT?
 - d. What are the challenges of CloudIoT? Discuss.

 5. **Attempt any two of the following:** 12
 - a. Is there any difference between compact city and smart city? Discuss.
 - b. Give various perspectives on definitions of smart city. What is the role of governance in development of smart city?
 - c. Discuss the Copenhagen zero carbon emission agenda and district of Songdo new resident integration.
 - d. Explain the fetal heart monitor and antenatal healthcare.
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